

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2023

Bachelor in Computer Applications

Course Title: **Database Management System**

Code No: **CACS255** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Define database and DBMS. Explain the benefits of DBMS in detail.
3. What are different SQL constraints to specify the rules for the data in the database? Explain.
4. What is relational algebra? Explain different RA operations.
5. How candidate key is different from primary key? Also explain foreign key in detail.
6. What is a stored procedure? How can it be used in a database? Explain with an example.
7. What are the major purposes of concurrency control? Explain two phase locking protocol in detail.
8. What is database security? Describe steps of query processing.

Group C

Attempt any TWO questions[2x10=20]

9. What are fully and partial functional dependencies? Explain 2NF and 3NF in detail.
10. How different join operations can be performed? Explain with proper examples. Consider a database of a shipment company. Shipped items are received into the system at a single retail center. Shipped items make their way to their destination via one or more standard transportation events (i.e. flights, truck deliveries). The transportation events follow a regular schedule. Create an ER diagram to represent this information.
11. Consider the following tables: Book(Bookid, Title, Publisher, Published Date, Price)Book_Authors (Bookid, Author name)Borrower (Card No, Name, Address, Contact)Bookissue (BookId, CardNo, IssuedDate, Due Date)Write SQL query for the following:a) Find the name of the books taken by Ramesh Sharma.b) Find the Issued book name and the borrower's name who have Issued date before 15th May, 2023 and due date after 15th June, 2023.c) Find the borrower's name who has taken the book written by Laxmi Prashad Devkota.d) Find the number of books Issued on the date 25th April, 2023.